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FedCoRe: Target-Adaptive Completion for Missing Modalities in Healthcare Federated Learning

Federated **Cro**ss-Modal **Re**presentation Completion

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Workshop on Distributed, Collaborative & Federated Learning
arxiv.org/abs/2608.18311 · nvidia.github.io/NVFlare/research/fedcore

PAPER

CODE

NVIDIA

THE SETTING

Complete-Modality Evaluation Is Optimistic for Federated Deployment

Hospitals differ in access to EHRs, chest radiographs, and ECGs — and patients within one hospital differ too. The hope: train better models from these uneven client inventories, so the resulting model still works when modalities are missing.

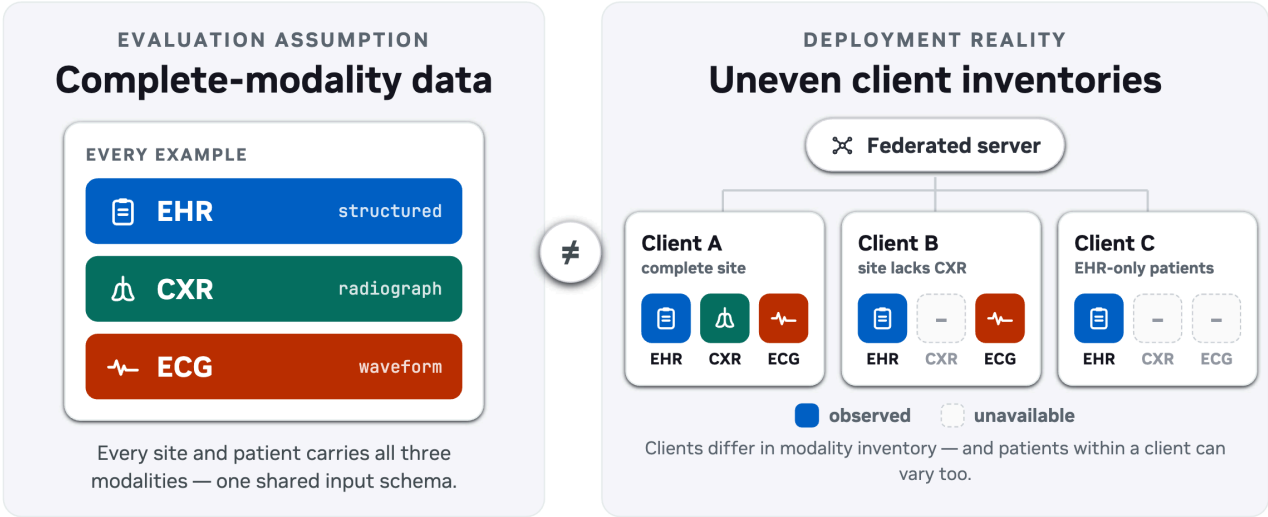


Fig. 1 — Evaluation assumption vs. deployment reality.

SCOPE

What “Completion” Means Here

IT IS

A **predictive correction** learned from paired modality-present / modality-removed examples — in representation space or on **classifier logits**.

IT IS NOT

Generation of a synthetic ECG or CXR — no modality is reconstructed.

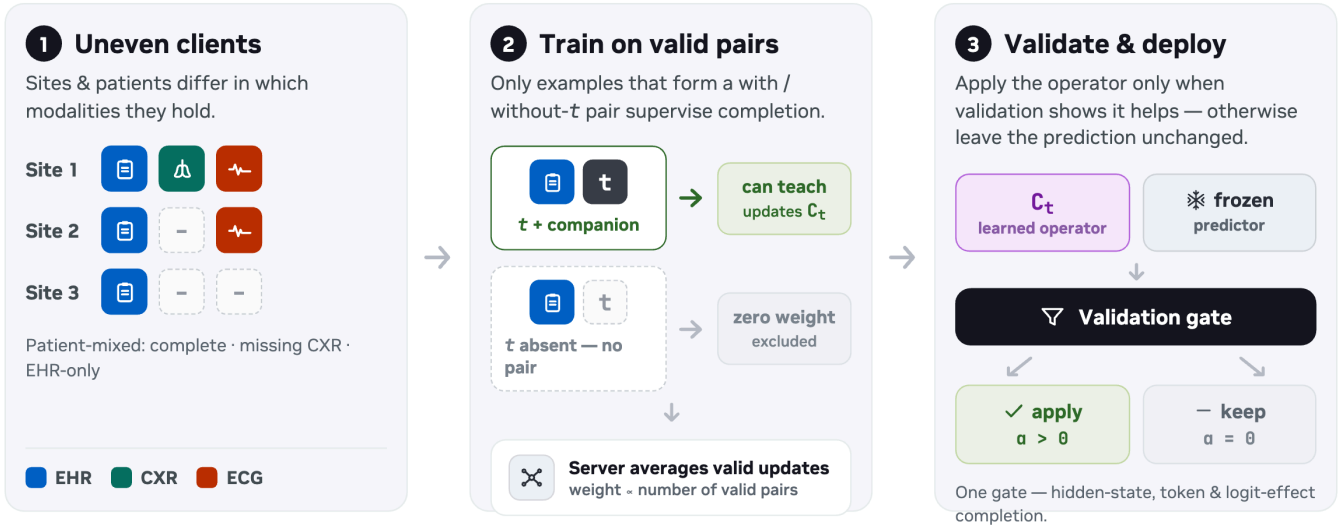
C1 **Paired-view supervision** — clients that observe a modality evaluate each example with and without it — no raw records shared.

C2 **Supervision-aware aggregation** — only clients with valid pairs update each operator, weighted by pair count.

C3 **Validation-gated deployment** — completion applies only when validation shows it helps — otherwise the prediction is unchanged.

METHOD

Paired Views Train It, Supervision Weights It, Validation Deploys It



1 · PAIRED VIEWS (2)

$h^{+t}_i = h_i(M_i), \quad h^{-t}_i = h_i(M_i \setminus \{t\})$

Valid only if $t \in M_i$ and $M_i \setminus \{t\} \neq \varnothing$ — same pair on the logits.

2 · AGGREGATION BY VALID PAIRS (7)

$\text{weight}_k \propto |P_{k,t}|, \quad |P_{k,t}| = 0 \Rightarrow \text{weight } 0$

One FedAvg round per modality; only operator weights cross the network.

3 · VALIDATION GATE (8)

apply c^* if $V^t_{\text{miss}}(c^*) \geq V^t_{\text{miss}}(c_0)$, else $\alpha = 0$

Candidates (source, α, τ) from a grid that always includes $\alpha = 0$; optional safety view may drop by at most ε.

METHOD · THE OPERATOR

The Completion Operator Is a Residual

C_θ adds a learned correction to the t-removed output; α_t sets the strength and $\alpha_t = 0$ leaves the prediction unchanged.

REPRESENTATION SPACE (3)

$$\tilde{h}^{t}_{i} = h^{-t}_{i} + \alpha_t \, C_\theta(h^{-t}_{i})$$

Residual on the pooled hidden state.

CLASSIFIER-LOGIT SPACE — HEADLINE RESULTS (4)

$$\tilde{\ell}^{t}_{i} = \ell^{-t}_{i} + \alpha_t \, C_\theta(h^{-t}_{i})$$

Residual on the frozen classifier output $\ell_i = g_\psi(h_i)$.

LOCAL OBJECTIVE

$$L_{\text{logit}} = \lambda_{\text{task}} \, \text{CE}(\tilde{\ell}^{t}_{i}, y_i) + \lambda_{\text{logit}} \, \| \tilde{\ell}^{t}_{i} - \ell^{+t}_{i} \|_2^2$$

Label term plus alignment with the t-present logits. For CXR, an instance gate scales the correction per example.

Encoder and head stay fixed — measured differences come from completion alone. One operator per modality; source, strength, and gate threshold are validation-selected.

BENCHMARK

A MIMIC-Derived Respiratory Deterioration Task

13,914
encounters
11,331 patients

9,777
train / 2,056 val
2,081 test

2.2%
positive class
304 of 13,914

4
simulated FL clients
patient-disjoint splits

EHR · MIMIC-IV

Structured encounter context, strictly pre-index.

CXR · MIMIC-CXR-JPG



Frontal chest radiograph, matched within 6 h.

ECG · MIMIC-IV-ECG



Lead II, 10 s.

LABEL AND LEAKAGE CONTROLS

- Invasive ventilation, ICU transfer, or in-hospital death within 48 h of the ECG–CXR index time.
- EHR context strictly pre-index; outcomes strictly post-index; ICU-at-index encounters excluded; ECG and CXR matched within 6 h.

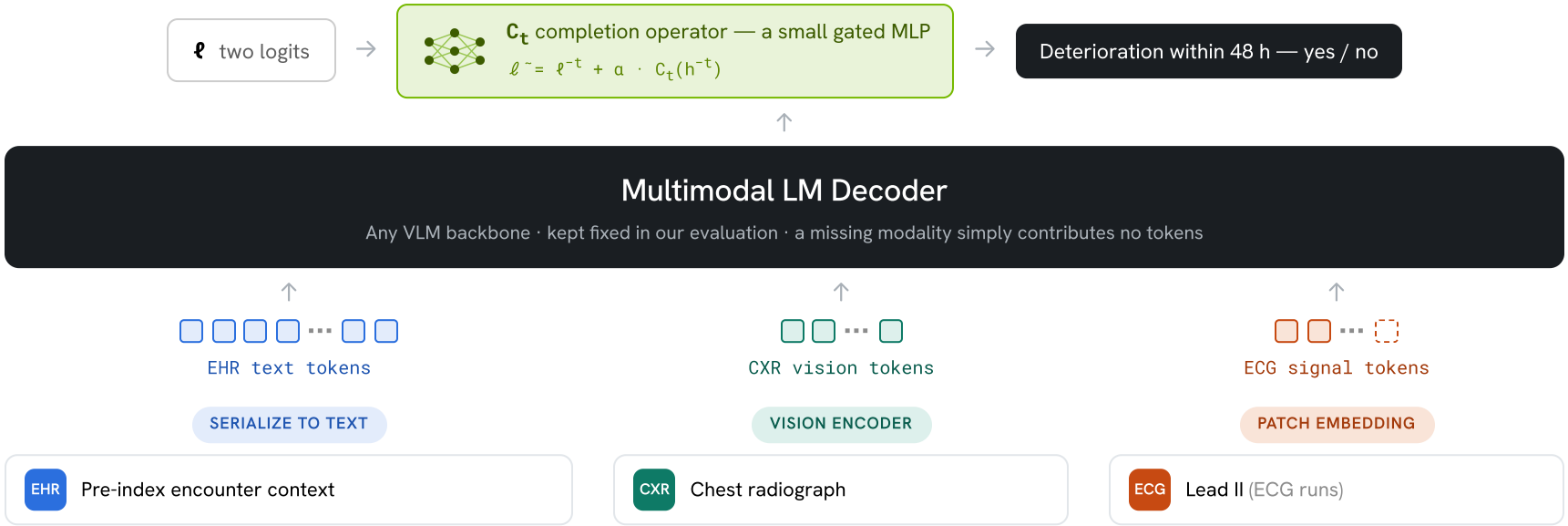
HETEROGENEITY TEMPLATES

	TEST PATIENTS MISSING ECG
No full-modality client negative control	66.3%
Patient-mixed, any-missing main heterogeneous setting	30.0%
Patient-mixed, heavy stress setting	36.3%

Expected test-set patient mass under each client mask; four clients per template.

BENCHMARK · PREDICTOR

A Multimodal VLM Predicts Deterioration



The interface is backbone-agnostic: completion needs only paired `t`-present / `t`-removed logits or representations — any multimodal predictor with that property plugs in.

RESULTS · PREDICTOR FROZEN

FedCoRe Recovers About Half the Lost AUROC

Test AUROC with the target modality hidden from the frozen predictor.

MISSING ECG · POOLED PATIENT-MIXED

49% OF LOST AUROC RECOVERED

ECG PRESENT	ECG HIDDEN	FEDCORE
0.6627	0.5781	0.6195

Hiding ECG costs 0.085 AUROC; FedCoRe returns +0.0415 [0.0023, 0.0778].

MISSING CXR · QWEN3-VL, 5 SEEDS

53% OF LOST AUROC RECOVERED

CXR PRESENT	CXR HIDDEN	FEDCORE
0.7042	0.5570	0.6440

Hiding CXR costs 0.147 AUROC; FedCoRe returns +0.087 on average, positive in 4 of 5 seeds.

$\alpha = 0$ If no correction terms can be supervised, the validation gate chooses $\alpha = 0$ and the prediction is unchanged.

CONCLUSION

FedCoRe — Federated Cross-Modal Representation Completion

- Modality-specific **predictive corrections** learned from paired views, aggregated only from clients that can supervise, applied only where validation supports them.
- Limits: simulated four-client masks; 42 test positives.
- Future work: validation on additional multimodal datasets and comparison against other approaches to handling missing modalities.

QUESTIONS WELCOME

Thank You

Implementation	Federated experiments run with **NVIDIA FLARE** · nvidia.github.io/NVFlare/research/fedcore
Data	MIMIC-IV, MIMIC-CXR, MIMIC-CXR-JPG, MIMIC-IV-ECG via PhysioNet
Paper	arxiv.org/abs/2608.18311
Contact	hroth@nvidia.com · Holger R. Roth, Ziyue Xu, Peter Cnudde

PAPER

CODE



When CXR Is Truly Absent, Validation Keeps the Baseline

The client modality mask **excludes CXR**, so no client can form a with/without pair — there is no paired reference to learn the correction from.

- Every candidate grid contains $\alpha = 0$; the safety constraint protects the observed view.
- Declining is empirical, not structural — no candidate cleared the bar on validation.
- Paired-example FedAvg transfers part of the effect in controlled rows (8/12, mean +0.034) — not in the deployment set.

DEPLOYMENT SIMULATION — CXR TRULY ABSENT

$\alpha = 0$

COMPLETION SWITCHED OFF

The client mask excludes CXR, so no paired reference exists. Validation finds no candidate that clears the bar and keeps the no-completion baseline.

SOURCE AND NEGATIVE CONTROLS

Server-selected sources +0.017 (2/6) · source-subset operators 0/12 · scalar recalibration: no lift · pseudo-CXR tokens rejected; forced, −0.082 AUROC.

Prior Multimodal FL Reconstructs, Retrieves, or Transfers the Missing Modality

Modality heterogeneity is a recognized central barrier in multimodal FL [18].

Pseudo-modality generation
Yan et al. [20]

Cross-contrast synthesis
FedMed-GAN [19] · Dalmaz et al. [3]

Cross-modal retrieval
CAR-MFL [15]

Contrastive ensembles
Yu et al. [21]

Embedding transfer
FedMEKT [13]

Prototype/mask completion
Bao et al. [1]

FedCoRe instead learns modality-specific corrections from paired outputs — no universal modality availability required.

Raw records stay local; no formal privacy guarantees are claimed — updates may leak without secure aggregation or differential privacy [12].

Paired Views Supervise a Residual Correction

A client holding the target modality t evaluates one example twice — with t, and with t removed. The difference is the training signal.

PAIRED VIEWS (2)

$$h^{+t}_i = h_i(M_i), \quad h^{-t}_i = h_i(M_i \setminus \{t\})$$

Valid only if $t \in M_i$ and $M_i \setminus \{t\} \neq \varnothing$ — same pair at the logit level. No raw records are shared.

C_θ adds a learned residual — to the pooled hidden state, or directly to the frozen classifier logits. α_t sets the strength; $\alpha_t = 0$ leaves the prediction unchanged.

REPRESENTATION SPACE (3)

$$\tilde{h}^{t}_i = h^{-t}_i + \alpha_t \, C_\theta(h^{-t}_i)$$

CLASSIFIER-LOGIT SPACE (4)

$$\tilde{\ell}^{t}_i = \ell^{-t}_i + \alpha_t \, C_\theta(h^{-t}_i)$$

LOCAL OBJECTIVE BEHIND THE HEADLINE RESULTS — CLASSIFIER-LOGIT COMPLETION

$$L_{\text{logit}} = \lambda_{\text{task}} \, \text{CE}(\tilde{\ell}^{t}_i, y_i) + \lambda_{\text{logit}} \, \| \tilde{\ell}^{t}_i - \ell^{+t}_i \|_2^2$$

Task loss plus alignment with the t-present logits — task-aware, not a reconstruction of the hidden state. CXR adds an effect-aware instance gate.

Encoder and head stay fixed — measured differences come from completion alone. One operator per modality; source, strength, and gate threshold are validation-selected.

Only Clients With Valid Pairs Contribute Updates

Only completion-operator updates cross the network; raw records stay local.

FEDAVG, WEIGHTED BY VALID PAIRS (7)

$$\theta^{r+1}_{t} = \sum_{k} \frac{|P_{k,t}|}{\sum_{j} |P_{j,t}|} \theta^{r+1}_{k,t}$$

Clients with no valid pairs receive zero weight — they cannot dilute or corrupt the global operator.

Aggregation granularity	one operator per modality *t*
Main ECG run	1 FedAvg round · 4 simulated clients

ONE ROUND, PER MODALITY *T*

Client 1	$\lvert P_{1,t} \rvert$ valid pairs	weight ∝ $\lvert P_{1,t} \rvert$
Client 2	$\lvert P_{2,t} \rvert$ valid pairs	weight ∝ $\lvert P_{2,t} \rvert$
Client 3	never observes *t* — no pairs	weight = 0

Validation-Constrained Selection Determines Deployment (8)

Keep the best candidate that helps where the modality is missing and does not hurt where it is present — otherwise change nothing.

1 · CANDIDATES

$c = (s, \alpha, \tau)$

source (FedAvg-aggregated or local) · completion strength · gate threshold

`α = 0 always in the grid`

2 · TWO VALIDATION TESTS

Helps where it matters — $V^t_{\text{miss}}(c) \geq V^t_{\text{miss}}(c_0)$ on validation cases missing t

Does no harm (optional) — a protected view (e.g. complete-modality cases) may drop by at most ε

3A · APPLY

$\alpha > 0$

best qualifying candidate

3B · KEEP BASELINE

$\alpha = 0$

prediction left unchanged

SAME RULE, OPPOSITE DECISIONS — MISSING ECG

Patient-mixed clients — pairs exist; a candidate clears the tests.

`0.5781` → `0.7235` AUROC

No full-modality client — no correction clears the bar.

`0.5781` → `0.5781` unchanged

Local Objective and the Effect-Aware CXR Interface

GENERAL OBJECTIVE — REPRESENTATION-SPACE COMPLETION (5)

$$L_i^t = \lambda_{\text{task}}\,\mathrm{CE}(g_\psi(\tilde{h}_i^t), y_i) + \lambda_{\text{logit}}\,\|g_\psi(\tilde{h}_i^t) - g_\psi(h^{+t}_i)\|_2^2 + \lambda_{\text{rep}}\,\|\tilde{h}_i^t - h^{+t}_i\|_2^2$$

λ_{task} task loss · λ_{logit} alignment with t-present logits · λ_{rep} optional representation consistency.

OBJECTIVE BEHIND THE HEADLINE RESULTS — CLASSIFIER-LOGIT COMPLETION

$$L_{\text{logit}} = \lambda_{\text{task}}\,\mathrm{CE}(\tilde{\ell}_i^t, y_i) + \lambda_{\text{logit}}\,\|\tilde{\ell}_i^t - \ell^{+t}_i\|_2^2$$

Both headline runs set $\lambda_{\text{rep}} = 0$; CXR adds effect-matching, probability-consistency, and gate losses.

EFFECT-AWARE CXR INTERFACE

$$d^{\text{CXR}}_i = \ell_i(M_i) - \ell_i(M_i \setminus \{\text{CXR}\}),\quad \Delta^{\text{CXR}}_i = \gamma_i\, r_\theta(h^{-\text{CXR}}_i),\quad \tilde{\ell}^{\text{CXR}}_i = \ell^{-\text{CXR}}_i + \alpha_{\text{CXR}}\,\Delta^{\text{CXR}}_i$$

Instance gate $\gamma_i \in [0, 1]$; losses add Smooth-L1 effect matching (λ_d), probability consistency (λ_p), gate supervision (λ_g), and gate sparsity (λ_s).

FL Heterogeneity Templates

Four clients each; missing-ECG mass is expected test-set patient mass under the client mask.

Template	Full-modality clients	Missing-ECG mass	Paired-example structure	Purpose
No full-modality client	0	66.3%	ECG-present but no complete site	Negative control
Patient-mixed any-missing	Mixed	30.0%	Complete and ECG-present patients within clients	Main heterogeneous setting
Patient-mixed heavy	Mixed	36.3%	Heavier missing-ECG patient mix	Stress setting

Frozen-Predictor Missing-ECG Test

Both methods select scale on validation; “Val-selected” also selects source. CIs use shared patient-level resamples across the six matched patient-mixed predictions.

Scenario	Completion	Seeds	Before	After	Full ref	Δ AUROC [95% CI]	AUPRC	Pos.
No full-modality client	FedAvg	7,17,42	0.5781	0.5781	0.6627	+0.0000 [0.0000, 0.0000]	0.0375 / 0.0375	0/3
No full-modality client	Val-selected	7,17,42	0.5781	0.5781	0.6627	+0.0000 [0.0000, 0.0000]	0.0375 / 0.0375	0/3
Patient-mixed combined	FedAvg	6 rows	0.5781	0.6195	0.6627	+0.0415 [0.0023, 0.0778]	0.0375 / 0.0368	6/6
Patient-mixed combined	Val-selected	6 rows	0.5781	0.7235	0.6627	+0.1455 [0.0617, 0.2213]	0.0375 / 0.0495	6/6

Reference-normalized gain: 0.0% · 0.0% · 49.0% · 171.8%. Bootstrap intervals preserve matched patient predictions but condition on the selected operator.

Frozen-Predictor Missing-CXR Completion

Delta intervals are paired patient-level bootstrap 95% CIs.

Seed	Full AUROC	Missing	Completed	Δ AUROC [95% CI]	Gap rec.	AUPRC before/after	α
7	0.6673	0.5237	0.6454	+0.1217 [0.0005, 0.2330]	84.7%	0.0204 / 0.0409	0.50
17	0.6969	0.5933	0.6228	+0.0295 [-0.0520, 0.1104]	28.5%	0.0402 / 0.0331	0.75
23	0.7303	0.6187	0.6187	+0.0000 [0.0000, 0.0000]	0.0%	0.0297 / 0.0297	0.00
31	0.7072	0.5188	0.6594	+0.1405 [0.0301, 0.2464]	74.6%	0.0255 / 0.0345	0.10
42	0.7192	0.5307	0.6739	+0.1431 [-0.0037, 0.2826]	75.9%	0.0327 / 0.0611	0.10
Mean	**0.7042**	**0.5570**	**0.6440**	**+0.0870**	**52.8%**	**0.0297 / 0.0399**	–

Operators, Optimization, and Search Grids

ECG / EHR OPERATOR

Architecture	2-layer logit-delta MLP
Hidden / dropout	2048 · 0.3
LR / batch	1e-4 · 128
Local epochs / rounds	50 · 1 FedAvg round
Loss weights	$\lambda_{task}=1$ · $\lambda_{logit}=4$ · $\lambda_{rep}=0$
α grid	{0, 2, 4, 8, 12}
Backend	Qwen-family, lead-II ECG patch tokens

CXR OPERATOR

Predictor	Qwen3-VL-8B-Instruct, LoRA r=8
Architecture	effect-aware gated MLP
Hidden / dropout	4096 · 0.25
LR / batch / epochs	5e-5 · 128 · 80
Loss weights	$\lambda_{task}=1$ · $\lambda_{logit}=8$ · $\lambda_{rep}=0$ $\lambda_{d}=1.0$ · $\lambda_{p}=0.5$ · $\lambda_{g}=1.0$ · $\lambda_{s}=0.005$
α grid	{0, .1, .25, .5, .75, 1, 1.25, 1.5, 2, 3, 4, 6, 8}
Gate thresholds	{0, .1, .25, .5, .75}

Selection uses validation only; every grid includes α = 0. A CXR candidate must improve remove-CXR validation AUROC, show ≥ 0.02 full-minus-removed headroom, and lose no more than ε = 0.005 on the full view. Validation/test manifests: 2,056 / 2,081 examples with 44 / 42 positives.

Baselines, Negative Controls, and Ablations

COMPARED ARMS

- No completion (“Before”), local-only completion, paired-example FedAvg, validation-selected.
- Full-modality runs are a reference, not a centralized pooled or reimplemented prior-method comparison.
- Sources evaluated: 3, 11, and 8 in the no-full-modality, any-missing, and heavy settings.

CXR SOURCES AND ABLATIONS

CONTROLLED TARGET REMOVAL

Paired-example FedAvg	+0.034 · 8/12
Server-selected source	+0.017 · 2/6
Source-subset operators	0/12 positive

SCENARIO-MASKED DEPLOYMENT

Validation decision	α = 0 · baseline kept

FORCED NEGATIVE CONTROLS

Scalar logit recalibration	zero lift
Pseudo-CXR tokens (forced)	−0.082

Validation rejects pseudo-CXR tokens; −0.082 is the cost of forcing completion anyway.

Test predictions were saved during replay but summarized only after validation fixed the source, strength, gate threshold, or no-completion choice; test metrics were never used for selection.